Supporting Computational Biologists at EMBL through Open-Source Tools and Four Pillars: Training, Community, Infrastructure, and Information

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THE CONTEXT

The **European Molecular Biology Laboratory** (EMBL) is an intergovernmental organisation dedicated to basic research in molecular life sciences. EMBL is supported by 27 **member states**, operates across **six sites** in Europe and has an **international staff** of over 2000 people. EMBL's research context is characterized by dynamism and heterogeneity, reflecting the organization's commitment to five missions:

- advancing fundamental **research** in molecular biology,
- **training** scientists,
- developing **new instruments and methods**,
- actively **engaging with industry**,
- providing services to the **broader community**.

In addition, EMBL's **Open Science policy** promotes international collaboration and innovation by ensuring that research findings, tools, and data are openly accessible to the scientific community.

EMBL's missions and operations are supported by different entities at the institute. Among those, the **Bio-IT project** provides a model for how institutional support can empower community-driven initiatives in **computational biology**. In line with the Open Science framework and culture, the project's goal is to establish a community of bioinformaticians through four pillars: training, community, infrastructure, and information. The project is managed by staff, providing a stable foundation for community-driven initiatives.

THE CHALLENGE

The Bio-IT project at EMBL faces several challenges due to the distributed nature of the institute, the diverse backgrounds of its international staff, and the dynamic nature of the research context. However, these challenges also present opportunities for growth and innovation.

One major challenge has been the need to **adapt to remote work** during the COVID-19 pandemic, which has highlighted the importance of digital infrastructure and communication tools. To address this challenge, the Bio-IT project has leveraged existing technologies to facilitate collaboration and exchange of information across all six EMBL sites in Europe.

Another challenge has been to **synergize the knowledge** of experts from **different disciplines**, including biologists, chemists, physicists, mathematicians, as well as software developers. The Bio-IT project has established community-building initiatives such as meetups, coding clubs, and internal chats to encourage interaction and collaboration among staff.

The community building initiatives are particularly useful in addressing another specific challenge of the institute, namely the **high staff turnover**, resulting from internal regulations and EMBL's role in training individuals who then return to the member states with their newly acquired expertise. To ensure the **sustainability** of the project activities and in line with the Open Science policy of the institute, the Bio-IT project commits to the adoption of open source tools for all its functions, from computational resources to the chat system.

Standardization of the training process has also been a challenge. The Bio-IT project, together with other entities at EMBL, has addressed this by mapping the training needs within the institute, by providing a platform to continuously collect them, and by trying to timely respond to these needs with an appropriate training offer. Advertisement of these events and materials is also critical, and the project has created a range of resources, such as guides, intranet links, and a blog, to ensure comprehensive access to information.

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THE VISION

Overall, the Bio-IT project's comprehensive approach mixing training, community building, infrastructure maintenance and information dissemination has been instrumental in establishing a community of computational biologists at EMBL. By sharing our model with other research institutions, we hope to inspire the development of similar programs and initiatives that can support computational biology research on a larger scale.

OUR SOLUTION

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Training is an essential component of the Bio-IT project. Researchers learn from their peers through **open-source materials** and **practical workshops**, run in a **hybrid** format to allow participation across all six EMBL sites in Europe. The training materials are built in a way that makes them compatible with **self-learning** as well as easy adoption by different trainers, and are designed and maintained **collaboratively**. The Bio-IT project values **community** expertise and often recruits trainers among EMBL researchers and staff. Continuous discussions are held to explore ways to provide trainers with **recognition** and benefits for their contributions. The training program focuses on building skills in data analysis, programming, and bioinformatics tools. Detailed data about past courses (including the number of hours spent on each topics and links to training materials) is stored in a **catalogue** from which learners can request the repetition of past courses.

But training alone is not enough. Computational biology research(ers) also require access to an **infrastructure** of state-of-the-art **computational resources** and **data management tools**. In collaboration with other entities at EMBL, the Bio-IT project provides guidance to the use of high-performance computing clusters, cloud-based resources, data storage and management services. In addition, it directly maintains coding and project hosting platforms such as the internal **JupyterHub** and **GitLab** servers. The project also serves as an ideal **testing ground** for the implementation of new biocomputational technologies before they are officially adopted at the institutional level.

These and other initiatives are aimed at building a sense of **community**, critical for researchers to feel connected to their peers and find support when they need it. Community building is implemented in regular meetups, coding clubs, and communication tools. **Meetups** (Bio-IT and Data Sciences taskforces) are held once a month and offer researchers the opportunity to discuss technical challenges, influence institutional decisions, request support and organise joint activities. **Coding clubs** provide an informal environment to share useful programming tools and technologies. The project also promotes and maintains communication tools like the open-source Mattermost **chat**, where researchers can ask questions, share resources, and receive support from their peers. In addition, the project has a "**grassroots table**" of internal experts who are available to answer questions and provide guidance to their colleagues.

The last component of a successful community management is the effective dissemination of **information**. The Bio-IT website serves as a hub to this aim, collecting **guides** for available resources, tools and best practices to EMBL researchers, including dedicated **pages for newcomers** to the institute, characterised by high turnover of personnel. The information helps in ensuring continuity of the promoted initiatives and of the resources usage and maintenance. Community news and relevant infrastructure updates are announced through the Bio-IT blog and the newsletter published on the institute's intranet.

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